

are the product roadmap for many vendors. In automobiles where several ELUs are used for dedicated tasks, multi-core MCUs will be a suitable choice," shares Tan.

Communication bus and integrated peripherals. Enabling fast computing through integration of Ethernet, CAN and USB has increased the use of MCUs in a variety of applications. "Over the years, we have seen more and more integration of peripherals such as analogue-to-digital converters (ADCs), pulse-width modulators (PWMs), universal asynchronous receivers/transmitters (UARTs), USB, I²C and LCD drivers with the controllers and the MCU is becoming a system-on-a-chip for key applications," informs Tyagi.

"Flexible and application-specific peripherals like CCU6 and universal serial interface controller (USIC) are now features of new MCU offerings," shares Ravi Kishore Ivaturi, head-auto-motive and MCU sales and marketing, Infineon.

Cost-effectiveness. Electronic products are growing in features, while their prices are going down. "Today, technology of 130 nm down to 90 nm is available with us to offer cost-effective devices," informs Thapliyal. On the other hand, research is under way for lowering the CMOS process nodes for cost-effectiveness. Tan shares, "Integrating a combination of useful peripherals also helps in reducing the total bill of materials (BOM) of the product."

MCUs and Indian market

The electronics market in India is evolving at a fast pace, and the role played by microcontrollers (MCUs) in electronics design cannot be underestimated. "With more application areas emerging, there has been a consistent pressure on semiconductor companies to come up with products that effectively counter the performance and efficiency challenges of traditionally available choices in 4-bit, 8-bit and 16-bit MCU designs," cites Shailash

Thakurdesai, business development manager (AEC), Texas Instruments.

The Indian market has been focus of many vendors for the specific needs and high-volume applications of energy-efficient and low-cost solutions. Ivaturi says, "In the Indian market, the usage of microcontrollers can be broadly classified into four main segments: consumer, industrial, automotive and telecom."

Although India, with its current booming economy, increased awareness levels and improving infrastructure, presents significant potential for innovative devices, players will still have to scrutinise carefully and work on the challenges ahead.

Thakurdesai adds, "Unlike other markets, in order to extend our reach in the Indian market, we will have to come up with India-specific products that would not only be portable but also deliver high-quality healthcare at affordable prices."

The increased market focus on energy efficiency is opening more uses for microcontrollers as manufacturers make devices that are smarter about how they consume power. Duvenhage says that this can manifest in the form of more efficient motor control algorithms or very simple implementations using the system's timers and switches to turn the device off. Thapliyal shares, "As per iSuppli and WSTS reports, the revenue growth for MCUs in India from year 2008 to year 2009 was 45 per cent, reaching \$125 million (excluding smart cards). The growth is expected to remain accelerated at 43 per cent for the year 2010."

The main drivers for the increasing usage of MCUs in India are metering, inverters and UPS. Other growing segments are solar/lighting, POS terminals, security systems and automobiles (for power train, body electronics, comfort and convenience).

Growth challenges

The biggest challenge for the microcontroller market is that it's constantly evolving at a fast pace. With so many

Vendor profiles

STMicroelectronics

STMicroelectronics' 8-bit MCU portfolio (STM8) consists of three families: STM8S, STM8L and STM8A. While STM8S devices are best suited for industrial applications, ultra-low-power STM8L devices are suitable for metering, small medical equipment and other portable applications. STM8A is dedicated for automotive applications like car window control, body control module and anti-pinch system. These families are core-compatible.

In 32-bit MCU portfolio, the STM32 family is based on ARM Cortex-M core. It brings a complete 32-bit product range that combines high performance, low power and low-voltage operation. STM32F series addresses a wide range of applications from lowest price-sensitive designs to the computing-intensive, high-memory footprint. The recently launched STM32L is suitable for ultra-low-power applications like medical devices and portable players.

Texas Instruments

Texas Instruments (TI) has a versatile MCU portfolio that includes the ultra-low-power MSP430 family, C2000 family and Stellaris ARM Cortex M3 family of microcontrollers.

TI has a variety of products that cater to metering applications, which is one of the primary focus areas in India. Some of the important product lines for metering applications include 16-bit MSP430 and 32-bit ARM Cortex M3 microcontrollers. Most of these products come with ultra-low-power capabilities, high-end performance levels, high accuracy with low error margins, and great degree of integration across interfaces. Apart from the silicon offering, TI offers fully certified software stacks for ZigBee implementation, DLMS, etc and basic metering software helping the user from a time-to-market perspective.

MCUs are also used in conjunction with LED drivers, MOSFETs and PWM controllers to bring about higher efficiency and increased battery life. Texas Instruments' 32-bit Piccolo series is used for precise multi-string LED control and Stellaris Cortex M3 MCUs for remote connectivity options.

semiconductor companies offering different configurations and solutions, the success of every device in the family is put to question.

"The MCU market needs strong quality requirements and certifications for controlling the haywire launch of